# Curriculum Vitae

Noura Howell
Associate Professor
Center for Software Technology
University of Southern Denmark
Vejle Campus

nourahowell.com
sites.gatech.edu/futurefeelings
scholar.google.com/citations?user=urDLnnEAAAAJ

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|---|---|---|---|---|---|---|
| Full papers: | 36 total, | in CORE: A*: 11 | A: 13 | B: 6 | Citations: 3087 |
| Extended abstracts: | 22 total, | in CORE: A*: 6 | A: 12 | B: 1 | h-index: 22 |
| Other writings: | 13 total | | | | i10-index: 33 |

## ▷ EDUCATION

2020 School of Information, University of California, Berkeley
PhD in Information Management & Systems with a Designated Emphasis in New Media

2012 Olin College of Engineering
BSc in Engineering with a Concentration in Computing

2007-8 Mississippi State University
Gap year in pure math for fun—graph theory, group theory, topology—and music theory

## ▷ ACADEMIC & INDUSTRY ROLES

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| 2026- | University of Southern Denmark, Center for Software Technology<br>Maersk Mc-Kinney Moller Institute, Faculty of Engineering | Associate Professor |
| 2021-25 | Georgia Institute of Technology, Department of Digital Media<br>School of Literature Media & Communication, Faculty of Humanities | Tenure-Track Assistant Professor |
| 2020-21 | North Carolina State University, Department of Communication | Tenure-Track Assistant Professor |
| 2017 | Singapore University of Technology & Design, Augmented Human Lab | Visiting Researcher |
| 2016 | Google Advanced Technologies and Projects, Project Jacquard | PhD Student Collaborator |
| 2014 | Intel Labs, Project Galileo | Software Engineering Contractor |
| 2013 | MIT Media Lab, Fluid Interfaces Group, openFrameworks, C++, user studies | Research Assistant |
| 2012 | The Echo Nest, Full Stack Web Developer, a startup later bought by Spotify | Full Stack Web Developer |
| 2010 | Microsoft | Program Manager Intern |
| 2007 | US Army Corps of Engineers, parallelised 10K+ line FORTRAN model with OpenMP | Volunteer Programmer |
| 2006 | North Carolina State University, Biological and Agricultural Engineering | Volunteer in Wet Lab |

## ▷ GRANTS & AWARDS

| | | | |
|---|---|---|---|
| 2026 | PI | 6.763.800 kr. | **Participatory Design Futuring Fabulation of Sustainable Futures with Youth and Local Communities**<br>**Novo Nordisk Start Package Grant** |
| 2025 | PI | $10.000 (≈64.000 kr.) | Community-Engaged Environmental Data Displays for Understanding the Environmental Impact of AI<br>Georgia Institute of Technology: Undergraduate Sustainability Education Innovation with Co-PIs Joe Bozeman and Michael Nitsche, community partner Decatur Makers |

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| | PI | $10.000<br>(≈64.000 kr.) | Sustainable Thermoregulation Strategies for Climate Change Resilience<br>Georgia Institute of Technology: Institute for Matter and Systems Initiative Lead<br>with Co-PIs Joe Bozeman and Matthew Flavin |
| 2024 | PI | $656.000<br>(≈4.212.000 kr.) | **Critically Reimagining Emotion AI by Combining AI Literacy & Design Futuring**<br>**NSF CAREER (CISE IIS HCC)** |
| | PI | $70.000<br>(≈449.000 kr.) | Applying Generative AI for STEAM Education: Supporting AI literacy and community engagement with marginalised youth<br>Georgia Institute of Technology: IDEAS – Institute for Data Engineering and Science<br>with Co-PI Michael Nitsche, community partner Decatur Makers |
| | PI | $10.000<br>(≈64.000 kr.) | Microscale Thermal Tech for Sustainability: Integrating Multidisciplinary Perspectives with Design Futuring Scenarios<br>Georgia Institute of Technology: Institute for Matter and Systems Initiative Lead<br>with Co-PI Joe Bozeman |
| | Co-PI | $10.000<br>(≈64.000 kr.) | Community-Engaged and Visceral Art Components for Smart and Sustainable Cities<br>Georgia Institute of Technology Undergraduate Sustainability Education Innovation Grant<br>with PI Joe Bozeman |
| | PI | $5.000<br>(≈32.000 kr.) | Generating Queer Histories: Prompting Critical Reflections on GenAI in the Archives<br>Atlanta Interdisciplinary AI Network<br>with PhD Candidate Alexandra Teixeira Riggs |
| 2023 | PI | $5.000<br>(≈32.000 kr.) | FuturesAtlanta: Creative Engagement with Generative AI Art as a Method for Designing Local Community Futures with Children in Atlanta<br>Atlanta Interdisciplinary AI Network<br>with PhD Student Supratim Pait |
| | PI | $10.000<br>(≈64.000 kr.) | High-Resolution, Fast-Switching, Non-Contact Thermal Interfaces<br>Ralph E. Powe Junior Faculty Enhancement Award |
| | PI | $6.000<br>(≈38.000 kr.) | Novel Thermal Interfaces<br>Georgia Institute of Technology COVID Faculty Relief Fund |
| | PI | $4.700<br>(≈30.000 kr.) | Towards Ethical, Inclusive Emotion AI Futures<br>Georgia Institute of Technology Small Grant for Research |
| 2022 | PI | $8.8K<br>(≈56.000 kr.) | PREMIER: Performance Residencies in Electronic Music for Interdisciplinary Education Research<br>Georgia Institute of Technology GVU/IPaT Community Engagement Grant<br>with Co-PI Alex Cohen |
| | Co-PI | $50K<br>(≈321.000 kr.) | Computational Craft Community Team Building<br>Georgia Institute of Technology Provost Funding<br>with PI Anne Sullivan, Co-PIs Vernelle Noel, Michael Nitsche, Sabetta Matsumoto |
| | PI | $5K<br>(≈32.000 kr.) | Exploring the Promise and Peril of Emotion AI<br>Georgia Institute of Technology Small Grant for Research |
| | PI | $6.5K<br>(≈41.000 kr.) | Emotion AI - Novel Facial Recognition Tech that Predicts Emotions - Investigating Sociotechnical Implications of Emotion AI<br>Georgia Institute of Technology COVID Faculty Relief Fund |

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| 2021 | PI | $10K (≈64.000 kr.) | **Engaging Diverse Students and Public Audiences with TensorFlow Emotion ML and Inflatable Architectural Immersive Environments**<br>with Co-PI Shawn Protz<br>**Google TensorFlow Faculty Award** |
| 2020 | Co-R | $4K (≈25.000 kr.) | An Alternate Lexicon for AI<br>with collaborators Noopur Raval and Co-PI Morgan Ames<br>Berkeley Center for Tech., Society, & Policy and Algorithmic Fairness & Opacity Group |
| 2019 | R | $25K (≈160.000 kr.) | Speculating 'Smart City' Cybersecurity with the Heart Sounds Bench: Détourning Data and Surveillance in Public Space<br>Berkeley Center for Long Term Cybersecurity |
| | Co-R | $5K (≈32.000 kr.) | Engaging Expert Stakeholders about the Future of Menstrual Biosensing Technology<br>with collaborators Sarah Fox, Richmond Wong, and Franchesca Spektor<br>Berkeley Center for Tech., Society, & Policy and Center for Long Term Cybersecurity |
| | R | $1.5K (≈9.600 kr.) | The Heart Sounds Bench<br>with mentee Stephanie Tang<br>Berkeley New Media Undergraduate Research Mentorship Award |
| 2018 | R | $4K (≈25.000 kr.) | Re-Centering the Body in Technological Utopias<br>with mentee Franchesca Spektor and mentor John Chuang<br>Berkeley Tech for Social Good |
| | R | $2K (≈12.000 kr.) | BaBench: Exploring Speculative Futures for Biosensing in Public Space<br>Berkeley Jacobs Innovation Center |
| | R | - | Berkeley Arts Research Center Fellowship<br>with mentor Greg Niemeyer |
| | Co-R | $5K (≈32.000 kr.) | Menstrual Biosensing Survival Guide<br>with collaborators Richmond Wong and Sarah Fox<br>Berkeley Center for Tech., Society, & Policy and Center for Long Term Cybersecurity |
| | R | $1.5K (≈9.000 kr.) | The Heart Sounds Bench<br>with mentee Victor Iancu<br>Berkeley New Media Undergraduate Research Mentorship Award |
| | R | - | Feeler Crawler Octopoets<br>with mentee Wenny Miao and mentor Greg Niemeyer<br>Berkeley New Media Summer Research Award |
| | R | $4K (≈25.000 kr.) | Berkeley Graduate Division Summer Grant |
| 2017 | S | - | Berkeley Outstanding Student Instructor Award |
| 2014 | S | $21K (≈13.000 kr.) | Berkeley Cota Robles Fellowship |

Roles:
PI: Principal Investigator, the main applicant and lead faculty on writing the proposal and doing the project, etc.
Co-PI: Part of a team of faculty on writing the proposal and executing the project, etc.
R: As a PhD student at the time, I could not officially PI grants, but I planned the research, wrote the grant proposals and budgets, did the research, wrote the research publications, and worked with finance admin. These grants came out of my research agenda, while faculty mentors signed off and gave high level advice.
Co-R: Part of a team of researchers as a PhD student, collaboratively responsible for writing the proposal, doing the project, etc.
S: Student recipient. Faculty applied for the award on my behalf.

## ▷ TEACHING

Interaction Design 2026
Lead instructor, SDU. Tangible fabrication, software design.

Principles of Interaction Design 2022-25
Lead instructor, Georgia Tech. Designing and evaluating screen-based interfaces. Contextual inquiry, task analysis, accessibility audit, heuristic evaluations.

Computer as Expressive Medium 2022-25
Lead instructor, Georgia Tech. Creative coding in p5.js for interactive media art, plus critical analysis of digital art.

Introduction to Computational Media 2025
Lead instructor, Georgia Tech. Introductory overview of the BSc Computational Media program, including design, media studies, and programming.

Technologies of Representation 2025
Lead Instructor, Georgia Tech. Datafication, technological mediation, politics of representation, AI as representation.

Introduction to Media Studies 2025
Lead instructor, Georgia Tech. Analyze and interpret media artifacts including expressive and political aspects.

Discovery & Invention 2024
Research methods for HCI and design. I restructured the curriculum so students learn methods by doing them in semester-long projects for external clients.

MS Project Capstone 2023-24
Team-teaching new class guiding MS projects on research methods, IRB, portfolio review, etc.

Special Topics in HCI Research 2022-24
HCI and design research methods for graduate students.

Critical Making with Emotion AI 2021
Lead instructor, Georgia Tech. Rapsberry Pi, Python, Google Cloud Vision API, built interactive art installation with Emotion AI.

Biosensing Technologies in Everyday Life 2021
Lead instructor, NC State University. Readings, discussion, and student-driven projects on sociotechnical implications of biosensing technologies.

Critical Making & Design Futuring 2021
Lead instructor, NC State University. Readings, discussion, and student-driven design futuring projects on topics such as algorithmic oppression, facial recognition, biometric surveillance, content moderation, etc.

Creative Programming & Electronics 2018
Teaching assistant, UC Berkeley. Hands-on Arduino, p5.js, circuits, and soldering.

Tangible User Interfaces 2016-18
Teaching assistant, UC Berkeley. Lesson plans, leading labs, design theory, Arduino, circuits, soldering. Design critique and project mentorship.

Deconstructing Data Science 2016
Teaching assistant, UC Berkeley. Machine learning methods with critical social analysis. Python tutoring, project advising.

Creative Code Immersive 2014
Teaching assistant, Gray Area Foundation for the Arts. Arduino, Processing, circuits, and JavaScript.

Code 510 Web Development 2014-18
After-school coding club for marginalised Oakland youth.

## ▷ ADVISING

PhD Advisor, University of Southern Denmark

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|---|---|---|
| Simona Mancusi | 2026-29 | Fabulation with Youth on Local Green Transitions, exact topic TBD |
| Attila Tuscher | 2026-29 | Fabulation with Youth on Local Green Transitions, exact topic TBD |

PhD Advisor, Georgia Institute of Technology

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|---|---|---|
| Supratim Pait | 2024-29 | Generative AI Literacy and Co-Creation with Diverse Youth and Artists |
| Xingyu Li | 2023-28 | Critically Reconfiguring Emotion AI via Participatory & Artistic Interventions |
| Sosuke Ichihashi | 2022-27 | Resonant Atmospheres: Prototyping Embodied Perception Interactions |
| Alexandra Teixeira Riggs | 2021-26 | Queering Tangible Interaction Design in Archival Experiences |
| Sylvia Janicki | 2021-25 | Designing with Data across Embodied Ecologies via Crip/Disabled Perspectives |

PhD Committee: Participate in PhD exams and offer research mentorship, Georgia Institute of Technology

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|---|---|---|
| Atefeh Mahdavi Goloujeh | 2025-27 | Representing Stakeholder Value Conflicts for Responsible AI Development |
| Chengzhi Zhang | 2023-28 | Co-Creative AI with Tangible Embodied Prototyping |
| Inha Cha | 2023-28 | How Technology Workers Practice AI Ethics, Adoption, and Resistance |
| Rachel Donley | 2020-26 | Designing Interactive Technology for Escape Room Experiences |
| Sara Milkes Espinosa | 2020-25 | A Study Of Platform Labor In US-based Online Secondhand Clothing Markets |
| Yuchen Zhao | 2019-25 | BioXR: Enhancing Engagement with Biosensing and Immersive Media |

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| Shubhangi Gupta | 2020-24 | Good Enough Explanations: How Local Publics Understand and Explain Civic AI |
| Erin Truesdell | 2019-23 | Designing Controllers for Collaborative Play |

PhD Committee External Member: Participate in PhD exams and offer research mentorship

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| Mathilde Gouin | 2024-27 | Situated Multispecies Encounters: Wearable Technology for Ecological Awareness, Instituto Superior Técnico, University of Lisbon |
| Richard Zhang | 2021-27 | Assembling Strangeness: Theorizing and Practicing Ostranenie for the Design of AI Systems, Northwestern University, Chicago, US |

MFA Committee: Participate in MFA thesis evaluation and offer mentorship

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| Cheyenne Hendrickson | 2023-25 | Auscultation: Medicalisation, Interactive Sound Sculptures, Georgia State University |

PhD External Evaluator: Evaluate theses, examine during midway seminars, etc.

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| Camilo Sanchez | 2026 | Contesting technodeterminism: Foregrounding explicit anticipation in Human–Computer Interaction research, Aalto University |
| Louie Søs Meyer | 2026 | (Dis)embodied by Algorithms: Queer Explorations of ML as a Design Material in Close to Body Technologies, ITU Copenhagen |
| Yann Seznec | 2024 | Play/Destroy: Sonic Interactions for a Dying World, KTH Stockholm |
| Chiara di Lodovico | 2024 | Designing w. Ambiguity in Self-Tracking Wearables & Data, Politecnico di Milano |
| Gabriele Barzilai | 2024 | Raising Ethical Awareness through Multisensory Systems, Politecnico di Milano |
| Vipula Dissanayake | 2022 | Multimodal Emotion Recognition in the Wild using Unsupervised Representation Learning Techniques, University of Auckland |

MSc Advisor, Georgia Institute of Technology

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| Yuhan Hou | 2023-26 | Investigating Trust Perceptions of LLMs in Personal Advice-Seeking |
| Vyshnavi Namani | 2023-25 | Supporting Computer Science TAs with Workshops and LLM Student Roleplay |
| Annette Guan | 2023-25 | Children's AI Literacy for Smart & Sustainable Cities |
| Gabriela Buraglia | 2023-25 | Children's AI Literacy for Smart & Sustainable Cities (joint with above) |
| Ruoyi Liu | 2023-25 | Redefining Beauty Standards to Support Young Women's Self-Esteem |
| Huaiwen Lou | 2023-25 | Improving User Experiences of Digital Health Tools |
| Supratim Pait | 2022-24 | PlayFutures: Imagining Civic Futures with AI & Puppets |
| Kosha Bheda | 2022-24 | Fostering Trust in Emotion AI Mental Health Applications with Design Recommendations for Sensitive Data |
| Natalie Huertas | 2021-23 | Tangible Gratitude: Conveying Emotion in Digital Spaces through Handwriting<br>Spring 2023 Commencement Speaker |
| Jui Patel | 2021-23 | Transcend the Generation Gap with Tangible Remote Communication Designs |
| Gwendolyn Hostetter | 2021-23 | Earworm: Digital Solutions for Manage & Minimizing Stuck Song Syndrome |
| Michelle Ramirez | 2021-22 | Digital Ghost Stories: Tangible Interactive Design for Oral Histories of the First Women at Georgia Tech |

MSc Committee: Participate in MSc evaluation and offer research mentorship, Georgia Institute of Technology

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| Marie Ow | 2023-25 | Preserving Culture Through Everyday Objects: A Tangible Story of Asian American Cultural Losses |
| Jasmine Kaur | 2023-25 | Choreographing the Future: Exploring Social and Ethical Aspects of AI in Dance |
| Xuanyu Hao | 2023-25 | Outmersive Games Based on Kinship Embodiment in Virtual Reality |
| Zhiyong Kong | 2022-24 | FactGuard: Chrome Plugin to Help Users Identify AI Hallucinations |
| Yue Zhu | 2022-24 | Web Streaming for Zoos for Broader Awareness and Fundraising |
| Xinyue Zhang | 2022-24 | Bill Buddy: AI Tool to Help Users Track & Manage Healthcare Spending |
| Vaishali Jain | 2022-24 | Notices and Choices for the Internet of Emotional Things |
| Jialuo Yang | 2022-24 | Investigating Tacit Knowledge in Woodworking with Multi-Layered Sensing |
| Xuanhao Li | 2022-24 | Knitwear with Algorithmically-Generated Structure as Tangible Interface |
| Deboah Cho | 2021-23 | IdeaLab: Building a Creative Project Tracker for People with ADHD |

## ▷ ACADEMIC SERVICE

Service to Intellectual Community

2021, '25 **National Science Foundation of the US (NSF) Panelist**: Reviewing NSF proposals, group deliberation with other panelists and NSF Program Officers to recommend which proposals get funded and provide feedback

Track Chair

2025 Designing Interactive Systems (DIS) Pictorials
2023 Designing Interactive Systems (DIS) Workshops

Subcommittee Chair

2025 Human Factors in Computing Systems (CHI) Papers - Design Subcommittee

Associate Chair

2022-26 Human Factors in Computing Systems (CHI)
2019-26 Designing Interactive Systems (DIS)
2023 Tangible, Embedded, and Embodied Interaction (TEI)
2023 Academic Mindtrek Papers

Reviewer

2016-27 Human Factors in Computing Systems (CHI)
2024 NordiCHI (Nordic Conference on Computer-Human Interaction)
2017-23 Designing Interactive Systems (DIS)
2021-22 Transactions on Human-Computer Interaction (ToCHI)
2018-22 Tangible Embedded Embodied Interactions (TEI)
2018-22 NordiCHI
2022 India HCI
2019-21 Computer-Supported Cooperative Work and Social Computing (CSCW)
2021 Journal of Textile Design Research and Practice
2020 ACM Group
2017 Design Issues

Service to University

2026 Joint Teaching Committee across Engineering & Science Faculties for SDU Vejle curriculum
2026 Hiring Committee, Faculty of Engineering, SDU
2026 Lab Space Infrastructure, Furniture, Equipment, Supplies Committee for the new research lab at SDU Vejle
2026 Jelling Stones Tech-a-thon Organising Committee, SDU Representative
2023-26 Institute Graduate Curriculum Committee, Georgia Tech
2024-25 Hiring Committee: for 2 open-rank tenure-track or tenured faculty in Digital Media, Georgia Tech
2023-24 Undergraduate Curriculum Committee, Computational Media Program, Georgia Tech
2023-24 Community Committee, Georgia Tech Dept. of Digital Media
2022-23 Executive Committee, Georgia Tech School of Literature, Media, & Communication
2020-21 Grants Committee, NC State Dept. of Communication
2020-21 Peer Teaching Evaluations Standards & Scheduling, NC State Dept. of Communication
2019 Office Redesign Committee, UC Berkeley School of Information
2015-16 PhD Student Representative to the Faculty, UC Berkeley School of Information

## ▷ INVITED TALKS, PANELS, & HOSTING WORKSHOPS

Reimagining Creativity: a Conversation about AI, Data, and Robotics in Art. Panelist with Bojana Ginn, Samuel Thurman. Moderator Birney Robert. Georgia Tech, 2024.

The Role of Generative AI in Teaching and Research in LMC. Panelist with Richmond Wong, Ida Yoshinaga, Brian Magerko, Zita Hüsing, Mark Leibert, Andrew Nance. Organizers Jay Bolter and Yeqing Kong. Georgia Tech, 2024.

Critical AI Literacy for Children with Schools in India, Finland, and the USA. Grace Hopper Conference for Women in Computing. With Co-Presenter Sumita Sharma. Orlando, FL, US, 2023.

Emotion AI and Fabulation: Seeking Biopolitical Futures of Respectful Care. Guest lecture in the course Biosensory Computing in the School of Information, course instructor John Chuang. UC Berkeley, 2023.

Emotion AI and Fabulation: Seeking Biopolitical Futures of Respectful Care. **Keynote speaker at Nordic Fabulation Network workshop**. Umeå, Sweden, 2023.

Introduction to Collaborative Autoethnography. Invited talk in Querying Experience workshop, Novel Interfaces for Musical Expression (NIME) conference. Online and Mexico City, 2023.

ChatGPT, ethics, and education for K12 teachers. Talk for GoSTEAM conference. With Sumita Sharma. CEISMC, Georgia Tech, Atlanta, US, 2023.

STEM for Enacting Ethical Change lecture with Finland high school students. Online to Oulu, Finland, 2023.

Full Radius Dance film screening of *Extension of Self: a dance between human and digital*. Panelist. Georgia Tech, 2023.

Emotional Robots and Digital Hormones: Japanese Perspectives on Human-Robot Coexistence, Jennifer Robertson. Book talk respondent. In the talk series Emerging Technologies and the Future of the Humanities. Emory University, US, 2023.

Toward an Affirmative Biopolitics: Reimagining Biodata with Feeling and Fabulation. Talk with Women in Music Technology. Georgia Tech, US, 2023.

Artists-in-Residence Panel for artists of PREMIER, Media Arts, and Library Artists-in-Residence programs. Moderator. Georgia Tech, 2023.

Exploring Emotion AI Ethics by Designing Tangible, Embodied, Social, Emotional Experiences with Biodata. Ethics & Coffee talk series. Georgia Tech, US, 2023.

Artists-in-Residence of PREMIER, Media Arts, and Library Artists-in-Residence programs. Panelist for Meet the Artists Reception. Georgia Tech, US, 2023.

Think Tank on Artist Residencies. Moderator for IPaT Tuesday Think Tank. Georgia Tech, US, 2022.

Design Futuring. Led a two-day invited workshop at University of Oulu, Finland, 2022.

Towards an Affirmative Biopolitics: Reimagining Biodata with Feeling and Fabulation. Invited lecture at Aalto University, as part of the Critical AI & Crisis Interrogatives Seminar. Aalto University, Finland, 2022.

Women in AI Finland invited talk with middle schoolers on AI and AI careers. International School in Oulu, Finland, 2022.

**Radio interview with US National Public Radio (NPR) City Lights** with Lois Reitzes, with Birney Robert, about the Heart Sounds Bench (my work) being in the exhibit Extension of Self (curated by Birney Robert), 2022.

Community Conversation: Impact, Arts, and Technology. Invited panelist alongside Gabriel Kahane and Felipe Barral. Hosted by the Atlanta Opera and Georgia Tech Arts. Atlanta, US, 2022.

Designing for Emotional Meaning-Making with Data. Invited lecture at The Scholars' Lab, University of Virginia, US, 2022.

Exploring the Promise and Peril of Emotion AI, Designing for Emotional Meaning-Making with Data, and Imagining an Affirmative Biopolitics with Data. Invited lecture at the GVU Seminar Series, Georgia Tech, US, 2021.

Designing for Emotional Meaning-Making with Data: Imagining an Affirmative Biopolitics. Invited lecture at the Coffee & Viz Research Exchange, North Carolina State University, US, 2021.

Beyond Big Tech: Careers in Social Impact, Tech for Good, and Research. Panelist, UC Berkeley, 2021.

Diversity Admissions Panel, UC Berkeley, 2020.

Reimagining Cybersecurity through the Design of the Heart Sounds Bench, talk at Center for Long-Term Cybersecurity Research Exchange, Berkeley, US, 2019.

Emotional Biosensing, talk at InfoCamp Conference, Berkeley, US, 2019.

Emotional Biosensing, talk at Bay Area Signal Hackers, Pandora, Oakland, US, 2019.

A Case Study of Emotional Biosensing: Tensions of Data-Driven Reflection, for the Society for the Social Studies of Science (4S), Boston, US, 2017.

Design Thinking: From Idea to Innovation, workshop facilitator for tech industry executives with the Augmented Human Lab, Colombo, Sri Lanka, 2017.

Emotional Biosensing, guest lecture for the class Mind-Reading and Telepathy for Beginners and Intermediates at UC Berkeley, 2017.

Machine Learning Introduction, guest lecture for the class City Planning 101 at UC Berkeley, 2017.

Information vs. Interaction: A Case Study of Affective Computing, lecture for the class Deconstructing Data Science at UC Berkeley, 2016.

Rethinking Data with Emotion and Materiality, guest lecture for the class Sensors, Humans, Data, Apps at UC Berkeley, 2016.

Rethinking Data with Emotion and Materiality, lecture for the class Tangible User Interfaces at UC Berkeley, 2016.

# Publications

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|---|---|---|---|---|---|---|
| Full papers: | 36 total, | in CORE: A*: 11 | A: 13 | B: 6 | Citations: | 3087 |
| Extended abstracts: | 22 total, | in CORE: A*: 6 | A: 12 | B: 1 | h-index: | 22 |
| Other writings: | 13 total | | | | i10-index: | 33 |

## ▷ FULL PAPERS

2026 Informal Embodied Auditing: Exploring Facial Emotion AI (FEAI) through Community Workshops
X Li, A T Riggs, Z Dai, C B Farmer, K G Morrison, N Howell
*Human Factors in Computing Systems* (*CHI*). Acceptance rate: 25%

Reconfiguring through Ruptures: Material Reconfigurations and Un/Making as Tangible Tactics for Queering AI-Generated Histories
A T Riggs, N Howell
*Human Factors in Computing Systems* (*CHI*). **Honourable Mention Award.** Acceptance rate: 25%, award for top 5%

Toxic Speculations: A Crip Posthuman Fabulation of Living in a Permanently Polluted World
S Janicki, H Biggs, N Howell
*Human Factors in Computing Systems* (*CHI*)

Human or AI Fortune-Telling: Exploring Users' Expectations and Trust Formation in Relationship Advice-Seeking with Human vs. AI I-Ching Divination
Y Hou, J Yang, R Y Wong, N Howell
*Proceedings of the ACM on Human-Computer Interaction* (*CSCW*)

Into the Makerverse: Communal Tangible Making and Place-Based Futuring with AI
S Pait, S Janicki, Y Hou, C B Farmer, L LaGasse Kowalski, J Long, A T Alvarenga Medina, X Li, V Namani, L Lew, M Nitsche, N Howell
*Tangible, Embedded, Embodied Interaction* (*TEI*)

Community Futures: Hybrid and Situated Critical AI Literacy
S Pait, S Janicki, Y Hou, M Nitsche, N Howell
*Artificial Intelligence in Education* (*AIED*)

HeartSway: Exploring Biodata as Poetic Traces in Public Space
Z Huang, Z Guo, X Li, X Ma, N Howell
*ACM Designing Interactive Systems* (*DIS*)

Children Envision Future GenAI Chatbots that are Bounded, Helpful, and Safe
T L Hong, Y Lao, M Iwata, J Tanaka, S Sharma, E White, H Tiensuu, N Howell
*Interaction Design & Children* (*IDC*)

2025 Queer Archival Un/Making as Tangible Information Activism
A T Riggs, M Mosher, A Sullivan, N Howell
*Designing Interactive Systems* (*DIS*)

Queer/Crip Body Mapping: Expressing Dynamic Bodily Experiences with Data
A T Riggs, S Janicki, T Moesgen, N Howell, K A Cochrane
*Designing Interactive Systems* (*DIS*)

Towards Designing for Everyday Thermal Experiences
S Ichihashi, K Bheda, N Howell
*Tangible, Embedded, Embodied Interaction* (*TEI*)

Swell by Light: An Approachable Technique for Freeform Raised Textures
S Ichihashi, N Howell, HJ Oh
*Tangible, Embedded, Embodied Interaction* (*TEI*)

Mold Sounds: Queering Ecologies in Polyphonic Material Explorations
A T Riggs, M Nitsche, N Howell
*Tangible, Embedded, Embodied Interaction* (*TEI*)

2024 Reflective Design for Informal Participatory Algorithm Auditing: A Case Study with Emotion AI
N Howell, W Hartsoe, J Amin, V Namani
*Nordic Human-Computer Interaction Conference* (*NordiCHI*)

Hydroptical Thermal Feedback: Spatial Thermal Feedback Using Visible Lights and Water
S Ichihashi, M Inami, H-N Ho, N Howell
*User Interface Software and Technology* (*UIST*)

Designing an Archive of Feelings: Queering Tangible Interaction with Button Portraits
A T Riggs, S Janicki, N Howell, A Sullivan
*Human Factors in Computing Systems* (*CHI*). Acceptance rate: 26%.

Crip Reflections on Designing with Plants: Disability Theory, Chronic Illness, & More-than-Human Design
S Janicki, N Parvin, N Howell
*Designing Interactive Systems* (*DIS*). **Best Paper Award.** Acceptance rate: 27%, award for top 1%

Promoting Criticality with Design Futuring with Young Children
S Sharma, N Howell, L Ventä-Olkkonen, N Iivari, G Eden, H Hartikainen, M Kinnula, E Durall, M Nitsche, J Okkonen, S Pait, E Rubegni, W Sluis-Thiescheffer, L van der Velden, U S Varanasi
*Nordic Human-Computer Interaction Conference* (*NordiCHI*)

"Tuning in and listening to the current": Understanding Remote Ritual Practice in Sufi Communities
S Kozubaev, N Howell
*Designing Interactive Systems* (*DIS*). Acceptance rate: 27%.

Red [Redacted] Theatre: Queering Puzzle-Based Tangible Interaction Design
A T Riggs, R Donley, T M Gasque, N Howell, A Sullivan
*Designing Interactive Systems* (*DIS*). Acceptance rate: 23%

Sensing Bodies: Engaging Postcolonial Histories through More-than-Human Interactions
S Janicki, A T Riggs, N Howell, A Sullivan, N Parvin
*Tangible, Embedded, Embodied Interaction* (*TEI*)

2023 Fabulation as an Approach for Design Futuring
M L Juul Søndergaard, N Campo Woytuk, N Howell, V Tsaknaki, K Helms, T Jenkins, P Sanches
*Designing Interactive Systems* (*DIS*). Acceptance rate: 24%

2022 Diffraction-in-Action: Designerly Explorations of Agential Realism Through Lived Data
P Sanches, N Howell, V Tsaknaki, T Jenkins, K Helms
*Human Factors in Computing Systems* (*CHI*). **Honourable Mention Award.** Acceptance rate: 14%, award for top 5%

Fabulating Biodata Futures for Living and Knowing Together
V Tsaknaki, P Sanches, T Jenkins, N Howell, L Boer, A Bitzouni
*Designing Interactive Systems* (*DIS*). Acceptance rate: 30%

Button Portraits: Embodying Queer History with Interactive Wearable Artifacts
A T Riggs, N Howell, A Sullivan
*International Conference on Interactive Digital Storytelling* (*ICIDS*)

2021 Cracks in the Success Narrative: Rethinking Failure in Design Research through a Retrospective Trioethnography
N Howell, A Desjardins, S Fox
*Transactions on Computer-Human Interaction* (*TOCHI*)

2020 Expanding Modes of Reflection in Design Futuring
S Kozubaev, C Elsden, N Howell, M L Juul Søndergaard, N Merrill, B Schulte, R Y Wong
*Human Factors in Computing Systems* (*CHI*). Acceptance rate: 23%

2019 Life-Affirming Biosensing in Public: Sounding Heartbeats on a Red Bench
N Howell, G Niemeyer, K Ryokai
*Human Factors in Computing Systems* (*CHI*). Acceptance rate: 24%

Vivewell: Speculating Near-Future Menstrual Tracking through Current Data Practices
S Fox, N Howell, R Y Wong, F Spektor
*Designing Interactive Systems* (*DIS*). Acceptance rate: 25%

2018 Tensions of Data-Driven Reflection: A Case Study of Real-Time Emotional Biosensing
N Howell, L Devendorf, T Vega Gálvez, R Tian, K Ryokai
*Human Factors in Computing Systems* (*CHI*). Acceptance rate: 26%

Emotional Biosensing: Exploring Critical Alternatives
N Howell, J Chuang, A De Kosnik, G Niemeyer, K Ryokai
*Proceedings of the ACM on Human-Computer Interaction* (*CSCW*). Acceptance rate: 26%

Capturing, Representing, and Interacting with Laughter
K Ryokai, E Duran, N Howell, J Gillick, D Bamman
*Human Factors in Computing Systems* (*CHI*). Acceptance rate: 26%

2016 ″I don′t want to wear a screen″: Probing Perceptions of & Possibilities for Dynamic Displays on Clothing
L Devendorf, J Lo, N Howell, J L Lee, N-W Gong, M E Karagozler, S Fukuhara, I Poupyrev, E Paulos, K Ryokai
*Human Factors in Computing Systems* (*CHI*). **Best Paper Award**. This award is for the 1% of accepted papers.

2013 On the L(2,1)-Labelings of Amalgamations of Graphs
S Spence Adams, N Howell, N Karst, D Sakai Troxell, J Zhu
*Discrete Applied Mathematics*

2009 Effect of Microbial Pretreatment on Enzymatic Hydrolysis and Fermentation of Cotton Stalks for Ethanol Production
J Shi, R R Sharma-Shivappa, M Chinn, N Howell
*Biomass and Bioenergy*

## ▷ EXTENDED ABSTRACTS

2026 PlayFutures: Imagining Civic Futures with AI and Puppets
S Pait, S Sharma, A Frith, M Nitsche, N Howell
*ACM SIG on Computer Science Education* (*SIGCSE*) *Technical Symposium*

Designing for Defamiliarization with Thermal Painting: Exploring Experiences of Dynamic Warmth in Painters′ Creative Processes
S Pait, S Ichihashi, X Li, H Xu, N Howell
*Tangible, Embedded, Embodied Interaction* (*TEI*) *Extended Abstracts*

Sufi Poetry and Introspection with Generative Verse
S Kozubaev, N Howell
*Designing Interactive Systems* (*DIS*) - *forthcoming June 2026*

2025 Lost in translation: Researchers' reflections on writing in English for CHI
S Sharma, B Norouzi, E P Greenwood White, E Durall Gazulla, M Y K Yousufi, N Iivari, N Howell, P Klemettilä, S Shahid, S Sarcar, X Li
*Human Factors in Computing Systems Extended Abstracts* (*alt.chi*)

Designing for Rest: Rethinking Access for / from Chronic Illness
S Janicki, J de Pereda Banda, L A Romero, X Guo, N Howell, A Stangl, S R Harris
*Human Factors in Computing Systems Extended Abstracts* (*alt.chi*)

2024 What should we do with Emotion AI? Towards an agenda for the next 30 years
N Andalibi, L Stark, D McDuff, R Picard, J Gratch, N Howell
*Proceedings of the ACM on Human-Computer Interaction* (*CSCW*) *Extended Abstracts*

Queering/Cripping Technologies of Productivity
S Janicki, A T Riggs, N Howell, A Sullivan, A Stangle
*Human Factors in Computing Systems Extended Abstracts* (*alt.chi*)

Mapping Futures and Futuring in HCI/Design
T Jenkins, V Tsaknaki, N Howell, L Boer, R Y Wong, N Campo Woytuk, M L Juul Søndergaard
*Designing Interactive Systems* (*DIS*) *Extended Abstracts*

Advancing Creative Physical Computing Education: Designing, Sharing, and Taxonomizing Instructional Interventions
D Byrne, K DesPortes, N Howell, M Louw, S Sterman
*Designing Interactive Systems* (*DIS*) *Extended Abstracts*

2023 Designing with Biosignals: Challenges, Opportunities, and Future Directions for Integrating Physiological Signals in Human-Computer Interaction
E R Stepanova, J Desnoyers-Stewart, A Kitson, B E Riecke, A N Antle, A El Ali, J Frey, V Tsaknaki, N Howell
*Designing Interactive Systems* (*DIS*) *Extended Abstracts*

Towards Mutual Benefit: Reflecting on Artist Residencies as a Method for Collaboration in DIS
L Devendorf, L Buechley, N Howell, J Jacobs, H-L Kao, M Murer, D Rosner, N Ross, R Soden, J Tso, C Zheng
*Designing Interactive Systems* (*DIS*) *Extended Abstracts*

Fabulating Biodata Futures for Flourishing and Vibrant Worlds
T Jenkins, M L Juul Søndergaard, P Sanches, V Tsaknaki, N Campo Woytuk, N Howell, K Helms, L Boer, J Tucker.
*Design Research Society: Nordic Design Research Society* (*Nordes*) *Extended Abstracts*

2022 Design Futuring for Love, Friendship, and Kinships: Five Perspectives on Intimacy
S Sharma, B F Schulte, R Fatás Arana, N Howell, A Twigger Holroyd, G Eden
*Human Factors in Computing Systems Extended Abstracts* (*alt.chi*)

Feeling Air: Exploring Aesthetic and Material Qualities of Architectural Inflatables
N Howell, S Protz, J Byrd, M Castellanos, A Elkins, J Hall, M Holdsworth, L Mallikeshwaran Rajagopal Sambasivan, C Noel, O Osiberu, R Patel, D Scallan, A Uhrich, A Anupam, B Bosley, R Donley, S Milkes Espinosa, M Ramirez, S Nayak, A-T Tran, Y Jia, Y Wang
*Nordic Human-Computer Interaction Conference* (*NordiCHI*) *Extended Abstracts*

2021 Calling for a Plurality of Perspectives on Design Futuring: An Un-Manifesto
N Howell, B F Schulte, A Twigger Holroyd, R Fatás Arana, S Sharma, G Eden
*Human Factors in Computing Systems Extended Abstracts* (*alt.chi*)

2020 Teachable Machine: Approachable Web-Based Tool for Exploring Machine Learning Classification
M Carney, B Webster, I Alvarado, K Phillips, N Howell, J Griffith, J Jongejan, A Pitaru, A Chen
*Human Factors in Computing Systems* (*CHI*) *Extended Abstracts*

Challenges and Opportunities for Designing with Biodata as Material
V Tsaknaki, T Jenkins, L Boer, S Homewood, N Howell, P Sanches
*Nordic Human-Computer Interaction Conference* (*NordiCHI*) *Extended Abstracts*

2018 Doodle Daydream: An Interactive Display to Support Playful and Creative Interactions Between Coworkers
S Elvitigala, S W T Chen, N Howell, D J C Matthies, S Nanayakkara
*Proceedings of the Symposium on Spatial User Interaction*

2017 Celebrating Laughter: Capturing and Sharing Tangible Representations of Laughter
K Ryokai, E Duran, D Bseiso, N Howell, J W Jun
*Designing Interactive Systems* (*DIS*) *Extended Abstracts*. Acceptance rate: 22%

Interrogating Biosensing in Everyday Life
N Merrill, R Y Wong, N Howell, L Stark, L Leahu, D Nafus
*Designing Interactive Systems* (*DIS*) *Extended Abstracts*

2016 Biosignals as Social Cues: Ambiguity and Emotional Interpretation in Social Displays of Skin Conductance
N Howell, L Devendorf, R Tian, T Vega Gálvez, N-W Gong, I Poupyrev, E Paulos, K Ryokai
*Designing Interactive Systems* (*DIS*). Acceptance rate: 26%

Representation and Interpretation of Biosensing
N Howell
*Designing Interactive Systems* (*DIS*) *Doctoral Consortium*

## ▷ OTHER WRITINGS: WORKSHOP PAPERS & POLICY COMMENTS

2024 Reflecting on the Role of Embodiment in Sufi Zikr
N Howell, S Kozubaev
Position paper at workshop *Navigating Intersections of Religion/Spirituality & HCI, NordiCHI*

PlayFutures: Imagining Civic Futures with AI and Puppets
S Pait, S Sharma, A Frith, M Nitsche, N Howell
Position paper at workshop *Child-Centered AI Design, CHI*

2023 Fast-Switching Spatial Thermal Display Using Water and Visible Lights
S Ichihashi, M Inami, N Howell
Position paper at workshop *Smell, Taste, and Temperature Interfaces, CHI*

2022 Comments in response to the Federal Trade Commission's Advance Notice of Proposed Rulemaking (ANPR) on a Trade Regulation Rule on Commercial Surveillance and Data Security
R Y Wong, W Hartsoe, N Howell
Comment ID FTC-2022-0053-1100

Children are the Future of Emotion AI
N Howell
Position paper at workshop *Age Against the Machine: Designing Ethical AI for and with Children, NordiCHI*

Years, Illness, Wildfires, Pandemic: Time and 'External Factors' in Design Ideation Processes|
N Howell
Position paper at workshop *Time and Its Study in Design Ideation Processes, NordiCHI*

Exploring Architectural Inflatables and Emotion AI to Activate Affective Public Space
N Howell
Position paper at workshop workshop *Tangible Interaction for Wellbeing, CHI*

2020 Opacity as Stubborn, Resistant Uncertainty
N Howell
Position paper at workshop *Embracing Uncertainty in HCI, CHI*

2019 Heart Sounds as a Means of Giving Form to Critical Alternatives with Biosensory Data
N Howell
Position paper at workshop *Doing Things with Research through Design, CHI*

2018 Reconfiguring Desire & Data
N Howell, G Niemeyer
Position paper at workshop *Grand Visions for Post-Capitalist Human-Computer Interaction, CHI*

2017 Personal Reflection as Creative Practice in Collaboration with Biosensing Machines
N Howell
Position paper at workshop *Mixed-Initiative Creative Interfaces, CHI*

2016 Textiles as On-Body Interactive Surfaces
N Howell
Position paper at workshop *Digital Craftsmanship: HCI Takes on Technology as an Expressive Medium, DIS*

2015 Connecting Two Oakland Neighbourhoods: Surveillance and Self-Representation
N Howell
Position paper at workshop *Inviting Participation through IoT: Experiments and Performances in Public Spaces, Critical Alternatives Decennial Conference in Aarhus*

## ▷ ARTISTIC OUTPUTS

2024 Heart Sounds Bench
N Howell
*Atlanta Art Fair*, Atlanta, Georgia, US

2022 Heart Sounds Bench
N Howell
*Georgia Tech Library Art Gallery*, Atlanta, Georgia, US

Embodied Transductions
N Howell
*New Interfaces for Musical Expression* (*NIME*), online

Infrastructural Membranes
N Howell
*Ferst Arts Center*, Atlanta, Georgia, US

2021 Feeling Air: A Psycho-Speculative Inflatable Happening
N Howell, S Protz et al
*Black Mountain College Museum + Arts Center Annual Conference*, Asheville, North Carolina, US

2019 Heart Sounds Buckets
N Howell, S Tang, K Ryokai
*Worth Ryder Art Gallery*, Berkeley, California, US

2018 Salaam Participatory Peace Sculpture
N Howell, S Clark, S Mohan
*Figment Arts Festival*, Oakland, California, US

Ebb Color-Changing Fabric
L Devendorf, N Howell, et al.
*CITRIS Tech Museum*, Berkeley, California, US

2017 Salaam Participatory Peace Sculpture
N Howell, S Clark, S Mohan
*Islamophobia Conference*, Berkeley, California, US

## ▷ DISSERTATION

2020 Emotional Meaning Making with Data
N Howell
University of California, Berkeley
*Proquest*